

Studio 7

Aggregation

Objectives

To introduce storing lists of objects with aggregation.

Exercise 1

1. What is the difference between aggregation and composition?
2. You will create a sketch that will allow the user to click in the sketch window and place objects to plan out a garden. At this stage you will just use a fixed width circle to represent a plant. When the user clicks in the sketch window a circle is drawn centered around that x and y position. A class diagram has been provided on Moodle.

Download the Studio 7 Ex 1 template sketch from Moodle.
3. Implement the Plant class using the notes below:

The drawPlant() method will draw a circle centred around the x and y position of the plant in the plant's color.

The growPlant() method will make the size of the plant increase by 1 pixel.
4. Implement the Garden class using the notes below:

The constructor will create a new arraylist object.

The addPlant() method will add a new plant object using the values passed to the method to the arraylist.

The drawGarden() method will draw all of the plants in the garden.

The growGarden() method will choose a random plant in the garden and make it grow.
5. Uncomment the code in the sketch tab to test your classes.

Exercise 2 (Tutorial handin Exercise)

1. You are going to start on the basics of a tower defence game. The game will have a series of teddy bears which will move across the sketch window. A class diagram has been provided on Moodle.

Download the Studio 7 Ex 2 template sketch from Moodle, a file called "teddybearicon.jpg" has been added to the sketch for you.

2. Implement the TeddyBear class using the notes below:

The direction field will have 2 possible values. 1 means it will move towards the right and -1 means it will move towards the left.

The constructor will load the image data into the img object and store the x and y position in a PVector object. The direction field should be set to 1.

The drawBear() method will draw the teddy bear icon to the sketch window using the position as the top left hand corner of the image.

The getWidth() method will return back the width of the image inside the img object.

The moveBear() method will move the teddy bear to the right by the amount passed into the method.

Test your teddy bear class by uncommenting the code in the sketch tab and making sure the teddy bear moves across the sketch window, you will see that it keeps going when it is passed the right hand edge of the sketch window.

3. Modify the moveBear() method so that when changing the x position, the amount is multiplied by the direction value. If the direction value is -1 then the amount will be subtracted rather than added.

Then before changing the x position add the code that checks if the right hand edge of the bear is greater than the width of the sketch window, if it is then set the direction to -1.

Test your code and make sure the teddy bear bounces off the right hand edge.

4. Modify the `moveBear()` method so that it will also bounce off the left hand edge of the sketch window and go towards the right again.

Test your code to make sure the teddy bear bounces off both edges of the sketch window.

5. Modify the `moveBear()` method so that whenever the bear bounces off an edge it is moved down by the height of the image.

Test your code and make sure the bear moves down the sketch window whenever it bounces.

6. Implement the Wave class using the notes below:

The constructor will set the number of bears in a wave to the value passed in. It will create a new array list for the bear list object.

Using a loop it will add the correct number of bears to the array list with the first bear starting at position 0,0 and the next bear immediately to the right of the previous bear.

Hint: When shifting the x position in the loop add the width of the bear at index position 0 in the list.

The `drawWave()` method will draw all the teddy bears in the wave to the sketch window.

The `moveWave()` method will move all the teddy bears by the amount passed in.

Test your wave class by commenting and uncommenting the appropriate code in the sketch window.